



Semester II 2023/2024

Subject : PROBABILITY & STATISTICAL DATA ANALYSIS
(SECI 1143)
Task : Chapter 5 & Chapter 6 - **100 marks**
Due Date :

INSTRUCTION:

1. This is **GROUP** assignment. Please clearly write your member's **NAME & MATRIC NUMBER** in the front page of the submission.
 2. This assignment contributes to **5%** of overall course marks.
 3. Only **HANDWRITTEN** submission is accepted:
 - a. Submissions using any reporting or statistical tools (e.g.: MS Word, MS Excel, etc.,) will be **REJECTED**.
 - b. Make sure the submission is neatly written. Any submission with handwriting that is unreadable, will be **REJECTED**.
 - c. For answer that need to draw graphs, using graph paper is optional. You can use plain paper.
 - d. Round your answers to **THREE** decimal places.
 4. Submission **hardcopy**
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QUESTION 1 [8 MARKS]

A study is conducted to determine the average monthly expenditure of a particular demographic group on groceries. You want to estimate the population mean monthly expenditure and construct a confidence interval to provide a range within which the true population mean lies. You randomly select **15 households** from a specific neighborhood and record their monthly grocery expenditures (in RM). The data are as shown in Table 1:

Table 1: Households monthly expenses

Household	A	B	C	D	E
Expenses (RM)	345	420	390	410	380
Household	F	G	H	I	J
Expenses (RM)	400	430	360	375	390
Household	K	L	M	N	O
Expenses (RM)	405	415	400	385	395

- a) Calculate a point estimate of the population mean, μ .
[4 marks]
- b) Construct and interpret a 95% confidence interval for μ .
[4 marks]

QUESTION 2 [12 MARKS]

A bank manager wants to determine if the average waiting time for customers in the bank's queue is less than 5 minutes. The manager randomly samples 500 customer waiting times and finds that the sample mean waiting time is 4.6 minutes, with a known population standard deviation of 1.2 minutes. The manager decides to conduct a hypothesis test with a significance level of 0.01.

Answer the following questions based on the above scenario.

- a. State the null and alternative hypotheses. [2 marks]
- b. Calculate the appropriate test statistic. [5 marks]
- c. Determine the critical value(s). [1 mark]
- d. Make a decision about the null hypothesis. [2 mark]
- e. Provide an interpretation of the decision in the context of the problem. [2 marks]

QUESTION 3 [10 MARKS]

A software company gives job applicants a test of programming ability and the mean for that test has been 160 in the past. Twenty-five job applicants are randomly selected from a large university and the produce a mean score of 183 with standard deviation of 12.

Use a 0.05 level of significance to test whether the mean score for the students form this university is greater than 160.

QUESTION 4 [10 MARKS]

A manufacturer makes steel bars that are supposed to have a mean length of 50 cm. A retailer suspects that the bars are running too long. A sample of 38 bars are taken and their mean length is determined to be 51 cm.

Using a 1% level of significance, perform a hypothesis test to determine whether the population mean is greater than 50 cm. Assume the population standard deviation is 3.6 cm.

QUESTION 5 [10 MARKS]

An agency recruited men and women for a study of how many words people speak in a day. A random sample of 51 males resulted in a mean of 16,576 words and a standard deviation of 7871 words.

Use a 0.01 significance level to test the claim that the males have a standard deviation that is greater than the standard deviation of 7460 words for females.

QUESTION 6 [15 MARKS]

Table 2 shows student course evaluation scores for courses taught by female professors and male professors in a university.

Table 2: Course evaluation scores for female and male professors

Female	4.3	4.3	4.4	4.0	3.4	4.7	2.9	4.0	4.3	3.4	3.4	3.3				
Male	4.5	3.7	4.2	3.9	3.1	4.0	3.8	3.4	4.5	3.8	4.3	4.4	4.1	4.2	4.0	4.4

Does it appear to be a difference in evaluation scores of course evaluation taught by female and male professors? Assume that the variance for the two population is equal.

- Find the mean and standard deviation for both samples. [4 marks]
- Use a 0.05 significance level to test the claim that the two samples have different mean. [4 marks]
- What is your conclusion based on your result in (b)? [2 marks]
- Using the same data and 0.01 significance level, test the claim that the variance of the course evaluation scores for female professor is higher than the male professors. [5 marks]

QUESTION 7 [15 MARKS]

Listed in Table 3 are the heights (in inches) of mothers and their first daughters.

Table 3: Mothers' Height vs Daughters' Height

Height of mothers (inches)	68.0	60.0	61.0	63.5	69.0	64.0	69.0	64.0	63.5	66.0
Height of daughters (inches)	68.5	60.0	63.5	67.5	68.0	65.5	69.0	68.0	64.5	63.0

- Find the differences of the paired values, D .
[5 marks]
- Find the mean of the differences ($\bar{D}$) and standard deviation of the differences (S_D)
[4 marks]
- Use a 0.05 significance level to test the claim that there is no difference in heights between mothers and their first daughters.
[6 marks]

QUESTION 8 [20 MARKS]

The management of a bank is interested in understanding whether there is a significant association between customer satisfaction levels with different banking services and the customers' preferred method of communication for receiving updates from the bank. A survey was conducted among 430 randomly selected customers, and their responses were recorded.

Table 4 shows the frequencies of customer satisfaction levels (Satisfied, Neutral, Dissatisfied) with banking services and their preferred method of communication (Email, SMS, Phone Call):

Table 4: Customer satisfaction levels

Preferable method of communication	Satisfied	Neutral	Dissatisfied	Total
Email	90	80	30	200
SMS	70	60	20	150
Phone call	40	30	10	80
Total	200	170	60	430

Using the provided data:

- calculate the value for e_i , $[o_i - e_i]^2 / e_i$ and χ^2 .

Cell, i	Observed Count, o_i	Expected Count, e_i	$[o_i - e_i]^2 / e_i$
...			
....			
....			
...			
$\sum \chi^2$			

[10 marks]

- Conduct a chi-square test to determine whether there is a significant association

between customer satisfaction levels with banking services and their preferred method of communication for receiving updates from the bank. Use a significance level of 0.05.
[10 marks]

$$\begin{aligned}
 1. a) \bar{x} &= \frac{5900}{15} \\
 &= 393.333 \\
 s &= \sqrt{\frac{\sum (x - \bar{x})^2}{n-1}} \\
 &= \sqrt{\frac{7083.33}{15-1}} \\
 &= 22.493
 \end{aligned}$$

$$\begin{aligned}
 b) df &= 15-1 = 14 \\
 \alpha &= 0.05
 \end{aligned}$$

$$\begin{aligned}
 t_{0.025, 14} &= 2.145 \\
 \bar{x} \pm (t_{\text{critical value}}) \left(\frac{s}{\sqrt{n}} \right) \\
 &= 393.333 \pm (2.145) \left(\frac{22.493}{\sqrt{15}} \right) \\
 &= 393.333 \pm 2.457 \\
 &= (380.876, 405.790)
 \end{aligned}$$

$\therefore$ We are 95% confident that the true population mean monthly grocery expenditures lies between 380.876 and 405.790.

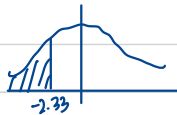
$$\begin{aligned}
 2. a) H_0: \mu &= 5 \\
 H_1: \mu &< 5
 \end{aligned}$$

$$\begin{aligned}
 b) \bar{x} &= 4.6 \\
 \sigma &= 1.2 \\
 n &= 500
 \end{aligned}$$

$$\begin{aligned}
 Z_{\bar{x}} &= \frac{\bar{x} - \mu_x}{\frac{\sigma}{\sqrt{n}}} \\
 &= \frac{4.6 - 5}{\frac{1.2}{\sqrt{500}}} \\
 &= -7.455
 \end{aligned}$$

$$c) \alpha = 0.01$$

$$Z_{0.01} = -2.330$$



d) Since $-7.455 < -2.330$, we reject the null hypothesis, H_0 .

e) There is sufficient evidence to support the claim that the average waiting time for customer in the bank's queue is less than 5 minutes.

3. $H_0: \mu = 160$

$H_1: \mu > 160$ (claim)

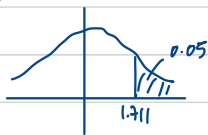
$n = 25$

$\bar{x} = 183$

$s = 12$

$\alpha = 0.05$ $df = 25 - 1 = 24$

$$t_0 = \frac{183 - 160}{\frac{12}{\sqrt{25}}} = 9.583$$



$t_{0.05, 24} = 1.711$

$\therefore$ Since $9.583 > 1.711$, we reject the null hypothesis, H_0 . There is sufficient evidence at the 0.05 level of significance to support the claim that the mean score for the students from this university is greater than 160.

4. $H_0: \mu = 50$

$H_1: \mu > 50$ (claim)

$n = 38$

$\bar{x} = 51$

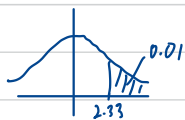
$\sigma = 3.6$

$\alpha = 0.01$

$$Z_{\bar{x}} = \frac{51 - 50}{\frac{3.6}{\sqrt{38}}}$$

$= 1.712$

$Z_{0.01} = 2.330$



$\therefore$ Since $1.712 < 2.330$, we fail to reject the null hypothesis, H_0 . There is not enough evidence at the 0.01 significance level to support the claim that the population mean length of the steel bars is greater than 50 cm.

5. σ_1 = s.d. for male

σ_2 = s.d. for female

$H_0: \sigma_1 = \sigma_2$

$H_1: \sigma_1 > \sigma_2$

$n_1 = 51$

$df_1 = 50$

$\bar{x}_1 = 16576$

$df_2 =$

$\sigma_1 = 7871$

$\sigma_2 = 7460$

$$F = \frac{(7871)^2}{(7460)^2} = 1.113$$

6. a) Female processors: ($n=12$)

$$\bar{x} = \frac{46.4}{12} \quad s = \sqrt{\frac{3.487}{12-1}}$$

$$= 3.867 \quad = 0.563$$

Male processors: ($n=16$)

$$\bar{x} = \frac{64.3}{16} \quad s = \sqrt{\frac{2.344}{16-1}}$$

$$= 4.019 \quad = 0.395$$

b) μ_1 = mean for female

μ_2 = mean for male

$$H_0: \mu_1 = \mu_2$$

$$H_1: \mu_1 \neq \mu_2 \text{ (claim)}$$

$$\alpha = 0.05$$

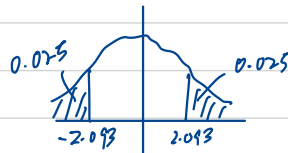
$$v = \frac{\left(\frac{0.563^2}{12} + \frac{0.395^2}{16}\right)^2}{\frac{\left(\frac{0.563^2}{12}\right)^2}{11} + \frac{\left(\frac{0.395^2}{16}\right)^2}{15}}$$

$$= 18.75 \approx 19$$

$$t_{0.025, 19} = 2.093$$

$$t_0^* = \frac{3.867 - 4.019}{\sqrt{\frac{0.563^2}{12} + \frac{0.395^2}{16}}}$$

$$= -0.799$$



$\therefore$ Since $-0.799 > -2.093$, we fail to reject the null hypothesis

c) There is no efficient evidence at the 0.05 significance level to support the claim that the two samples have different mean.

d) $H_0: \sigma_1^2 = \sigma_2^2$

$$H_1: \sigma_1^2 > \sigma_2^2$$

$$\alpha = 0.01$$

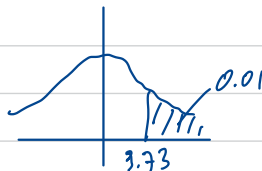
$$F = \frac{0.563^2}{0.395^2}$$

$$= 2.032$$

$$df_1 = 12 - 1 = 11$$

$$df_2 = 16 - 1 = 15$$

$$F_{0.01, 11, 15} = 3.73$$



$\therefore$ Since $2.032 < 3.73$, we fail to reject the null hypothesis. There is not enough evidence at the 0.01 significance level to support the claim that the variance of the course evaluation scores for the female processor is higher than the male professors.

7. a)

Height of mother	Height of daughters	Differences, D	D^2
68.0	68.5	-0.5	0.25
60.0	60.0	0.0	0.0
61.0	63.5	-2.5	6.25
63.5	67.5	-4.0	16
69.0	68.0	1.0	1
64.0	65.5	-1.5	2.25
69.0	69.0	0.0	0
64.0	68.0	-4.0	16
63.5	64.5	-1.0	1
66.0	63.0	3.0	9

$$\sum D = -9.5 \quad \sum D^2 = 51.75$$

b) $\bar{D} = \frac{-9.5}{10}$

$$= -0.95$$

$$s_D = \sqrt{\frac{51.75 - \frac{(-9.5)^2}{10}}{10-1}}$$

$$= 2.179$$

c) $H_0: \mu_D = 0$

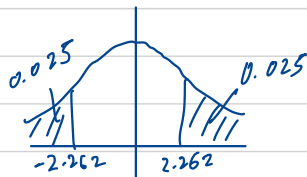
$$H_1: \mu_D \neq 0 \text{ (claim)}$$

$$\alpha = 0.05, \text{ df} = 10-1 = 9$$

$$t = \frac{-0.95 - 0}{\frac{2.179}{\sqrt{10}}}$$

$$= -1.379$$

$$t_{0.025, 9} = 2.262$$



$\therefore$ Since $-1.379 > -2.262$, we fail to reject the null hypothesis, H_0 . There is not enough evidence at the 0.05 significance level to support that there is no difference in heights between mothers and their first daughters.

$$4. H_0: \mu = 50$$

$$H_1: \mu > 50$$

$$Z = \frac{\bar{X} - \mu}{\frac{s}{\sqrt{n}}}$$

$$= \frac{51 - 50}{\frac{3.6}{\sqrt{38}}}$$

$$= 1.712$$

$$\alpha = 0.01$$

$$Z_{\alpha, n} = 2.330$$

$\therefore$ Since $1.712 < 2.330$, we fail to reject the null hypothesis, H_0 . There is not enough evidence at the 0.01 significance level to support the claim that the population mean length of the steel bars is greater than 50 cm.

$$5. H_0: \sigma = 7460$$

$$H_1: \sigma > 7460$$

$$\chi^2 = \frac{(n-1)s^2}{\sigma^2}$$

$$= \frac{(51-1)(7871)^2}{(7460)^2}$$

$$= 55.661$$

$$\alpha = 0.01$$

$$df = 51 - 1 = 50$$

$$\chi^2_{0.01, 50} = 76.154$$

$\therefore$ Since $55.661 < 76.154$, we fail to reject the null hypothesis, H_0 . There is not enough evidence at the 0.01 significance level to support the claim that the standard deviation of words spoken by males is greater than the standard deviation of 7460 words for females.

$$6a. \bar{X}_{female} = \frac{4.3 + 4.3 + 4.4 + 4.0 + 3.4 + 4.7 + 2.9 + 4.0 + 4.3 + 3.4 + 3.4 + 3.3}{12}$$

$$= 3.867$$

$$S^2_{female} = \frac{(4.3 - 3.867)^2 + (4.3 - 3.867)^2 + (4.4 - 3.867)^2 + (4.0 - 3.867)^2 + (3.4 - 3.867)^2 + (4.7 - 3.867)^2 + (2.9 - 3.867)^2 + (4.0 - 3.867)^2 + (4.3 - 3.867)^2 + (3.4 - 3.867)^2 + (3.4 - 3.867)^2}{12 - 1}$$

$$= 0.583$$

$$\bar{X}_{male} = \frac{4.5 + 3.7 + 4.2 + 3.9 + 3.1 + 4.0 + 3.8 + 3.4 + 4.5 + 3.8 + 4.3 + 4.4 + 4.1 + 4.2 + 4.0 + 4.4}{16}$$

$$= 4.019$$

$$S^2_{male} = \frac{(4.5 - 4.019)^2 + (3.7 - 4.019)^2 + (4.2 - 4.019)^2 + (3.9 - 4.019)^2 + (3.1 - 4.019)^2 + (4.0 - 4.019)^2 + (3.8 - 4.019)^2 + (3.4 - 4.019)^2 + (4.5 - 4.019)^2 + (3.8 - 4.019)^2 + (4.3 - 4.019)^2 + (4.4 - 4.019)^2 + (4.1 - 4.019)^2 + (4.2 - 4.019)^2 + (4.0 - 4.019)^2 + (4.4 - 4.019)^2}{16 - 1}$$

$$= 0.345$$

$$6b. H_0: \mu_1 = \mu_2 \text{ where } \mu_1 = \text{mean course evaluation scores for female professors.}$$

$$H_1: \mu_1 \neq \mu_2 \text{ } \mu_2 = \text{mean course evaluation scores for male professors.}$$

$$S_p^2 = \frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{n_1 + n_2 - 2}$$

$$= \frac{(12 - 1)(0.583)^2 + (16 - 1)(0.345)^2}{12 + 16 - 2}$$

$$= 0.224$$

$$S_p = \sqrt{0.224}$$

$$= 0.473$$

$$t = \frac{\bar{X}_1 - \bar{X}_2 - 0}{S_p \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}}$$

$$= \frac{3.867 - 4.019}{(0.473) \sqrt{\frac{1}{12} + \frac{1}{16}}}$$

$$= -0.841$$

$$\alpha = 0.05, \frac{\alpha}{2} = 0.025$$

$$df = 12 + 16 - 2 = 26$$

$$t_{\alpha/2, 26} = 2.056$$

$\therefore$ Since $-0.841 < 2.056$, we fail to reject the null hypothesis, H_0 . There is not enough evidence at the 0.05 significance level to support the claim that the two samples have different mean.

c. Based on the t-test result, $-0.841 < 2.056$, that means we fail to reject the null hypothesis, H_0 . There is not enough evidence at the 0.05 significance level to support the claim that the two samples have different mean.

6d. $H_0: \sigma_1^2 = \sigma_2^2$ where σ_1^2 = variance of the course evaluation scores for female professors.
 $H_1: \sigma_1^2 > \sigma_2^2$ σ_2^2 = variance of the course evaluation scores for male professors.

$$F = \frac{s_1^2}{s_2^2} = \frac{0.568^2}{0.595^2} = 2.032$$

$$\alpha = 0.01$$

Degree of freedom: Numerator = 12 - 1 = 11

Denominator = 16 - 1 = 15

$$F_{0.01, 11, 15} = 3.67$$

$\therefore$ Since $2.032 < 3.67$, we fail to reject the null hypothesis, H_0 . There is not enough evidence at the 0.01 significance level to support the claim that the variance of the course evaluation scores for female professors is higher than the male professors.

a) differences of the paired values, D

X_1	X_2	$D = X_1 - X_2$	$D^2 = (X_1 - X_2)^2$
68.0	68.5	-0.5	0.25
68.0	68.0	0	0
61.0	65.5	-4.5	20.25
63.5	67.5	-4.0	16.00
69.0	68.0	1.0	1.00
64.0	65.5	-1.5	2.25
69.0	69.0	0	0
69.0	68.0	1.0	1.00
63.5	64.5	-1.0	1.00
66.0	63.0	3.0	9.00
		$\Sigma D = -9.5$	$\Sigma D^2 = 51.75$

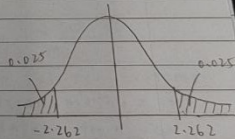
$$\bar{D} = \frac{\Sigma D}{n} = \frac{-9.5}{10} = -0.95$$

$$S_D = \sqrt{\frac{\Sigma D^2 - \frac{(\Sigma D)^2}{n}}{n-1}} = \sqrt{\frac{51.75 - \frac{(-9.5)^2}{10}}{10-1}} = 2.1788$$

$$H_0: \mu_D = 0 \quad \alpha = 0.05$$

$$H_1: \mu_D \neq 0 \quad df = 10 - 1 = 9$$

$$t = \frac{\bar{D} - \mu_D}{\frac{S_D}{\sqrt{n}}} = \frac{-0.95 - 0}{\frac{2.179}{\sqrt{10}}} = -1.379$$



$$t_{0.025, 9} = 2.262 = -2.262$$

$\therefore$ Conclusion: Since $-1.379 > -2.262$, We fail to reject H_0 . Thus, there is not enough evidence to claim that there is no difference in height between mother and their first daughters

Cell, i	Observed count, O_i	Expected count, E_i	$(O_i - E_i)^2 / E_i$
1,1	90	$(100)(200)/430 = 46.02$	0.098
1,2	80	$(170)(200)/430 = 79.07$	0.011
1,3	30	$(60)(200)/430 = 27.91$	0.157
2,1	70	$(200)(150)/430 = 69.77$	0.001
2,2	60	$(170)(150)/430 = 59.30$	0.008
2,3	20	$(60)(150)/430 = 20.93$	0.041
3,1	40	$(200)(80)/430 = 37.21$	0.209
3,2	30	$(170)(80)/430 = 31.63$	0.084
3,3	10	$(60)(80)/430 = 11.16$	0.121
$\Sigma \chi^2$			0.730

b) H_0 : No significant association

H_1 : There is significant association

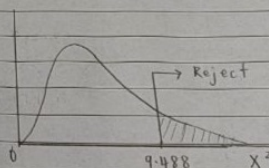
$$\alpha = 0.05$$

$$df = (3-1)(3-1) = 4$$

$$\chi^2_{0.05, 4} = 9.488$$

$$\text{Test statistic: } \chi^2 = 0.730$$

$$\text{Critical value: } \chi^2 = 9.488$$



Test statistic < critical value which is $0.730 < 9.488$, so we do not reject H_0 and conclude that there is no significant association between customer satisfaction levels and their preferred method of communication for receiving updates from the bank.